

Saeid Bayat

Postdoctoral Research Fellow

+1 (217) 819 7166
saeidb@umich.edu
www.saeidbayat.com
Google Scholar

Education

- 2024–Now **Research Fellow in Naval Architecture and Marine Engineering**, *University of Michigan*, Ann Arbor, USA
Advisor: Prof. Lei Zuo
- 2019–2024 **Ph.D. in Industrial and Enterprise Systems Engineering**, *University of Illinois at Urbana–Champaign*, Urbana–Champaign, USA
Advisor: Prof. James T. Allison
- 2015–2018 **M.Sc. in Mechanical Engineering (Major: Mechatronics)**, *Sharif University of Technology*, Tehran, Iran
- 2011–2015 **B.Sc. in Mechanical Engineering**, *Iran University of Science and Technology*, Tehran, Iran

Research Focus

- Marine renewable energy systems
- Control co-design
- Multidisciplinary design optimization
- AI-guided engineering design
- Data-driven modeling and surrogate modeling
- Dynamics and control
- Experimental validation and hardware-in-the-loop testing

Research Profile

- Publications **17** peer-reviewed journal papers (**13** first-author, **1** co-first-author) and **6** peer-reviewed conference papers.
- Career Stages M.Sc. (**2** journal), Ph.D. (**12** journal, **3** conference), Postdoc (**3** journal, **3** conference).
- Metrics 209 citations | h-index: 10 | i10-index: 10

Honors and Awards

- 2026 **Winner, Power at Sea Prize – DEVELOP Phase**, *U.S. Department of Energy (DOE) / National Laboratory of the Rockies (NLR)*, Award: \$75,000
Team AquaSync. Selected as a winner in the Power at Sea Prize DEVELOP Phase for the development of offshore renewable-energy technologies.

- 2025 **Best Paper Award Winner (1st place)**, *ASME IDETC/CIE 2025 VIB Track*, Anaheim, CA, USA
"Development of a Scaled Hydrofoil-Based Marine Energy Converter: Design, Modeling, and Parametric Optimization"
- 2018 **Selected Technologist Award**, *certificate of commendation from Sharif University of Technology*, Tehran, Iran
- 2015 **Direct admission (without entrance exam)**, *elite student admission for M.Sc. at Sharif University of Technology*, Tehran, Iran
- 2015 **Rank 5 in Bachelor's class**, *fifth among all 100 Mechanical Engineering graduates of Iran University of Science and Technology*, Tehran, Iran
- 2012, 2014 **Elected as an Elite Student**, *by the President of Iran University of Science and Technology*, Tehran, Iran
- 2011 **Ranked in the top 0.2%**, *among over 350,000 students in the nationwide bachelor's entrance exam, admitted to Iran University of Science and Technology*, Tehran, Iran

Journal Publications

- 2026 **Saeid Bayat**, Jerry Zuo, and Jing Sun. "Modeling and Dynamic Simulation of a Hybrid Floating Wind-Wave Platform with Integrated Flap-Type Wave Energy Converters." *Ocean Engineering*, [Link](#).
- 2026 **Saeid Bayat** and Lei Zuo. "Multidisciplinary Control Co-Design of a Spar-Torus Hybrid Wind-Wave Energy System." *Journal of Mechanical Design*, 1–14 (2025). [Link](#)
- 2026 **Saeid Bayat**, Chad Peterson, Yong Hoon Lee, Jenna Iori, and James T. Allison. "Advancing Wind Turbines Through Control Co-Design: An Integrative Review." *Applied Energy*, vol. 416, p. 127951, 2026. [Link](#).
- 2025 Vishnu Vijayasankar, **Saeid Bayat**, and Lei Zuo. "Development of a Scaled Hydrofoil-Based Marine Energy Converter: Design, Modeling, and Parametric Optimization." *Journal of Vibration and Acoustics*, 1–12 (2025). [Link](#)
- 2025 Yong Hoon Lee, **Saeid Bayat**, and James T. Allison. "Wind turbine control co-design using dynamic system derivative function surrogate model (DFSM) based on OpenFAST linearization." *Applied Energy*, 396:126203 (2025). [Link](#)
- 2025 **Saeid Bayat**, Yong Hoon Lee, and James T. Allison. "Nested control co-design of a spar buoy horizontal-axis floating offshore wind turbine." *Ocean Engineering*, 328:121037 (2025). [Link](#)
- 2025 **Saeid Bayat** and James T. Allison. "Impact of control strategies on the control co-design of spar floating offshore wind turbines." *Ocean Engineering*, 336:121763 (2025). [Link](#)
- 2025 Yong Hoon Lee, **Saeid Bayat**, James T. Allison, Md. Sanower Hossain, and D. Todd Griffith. "Multidisciplinary modeling and control co-design of a floating offshore vertical-axis wind turbine system." *Journal of Mechanical Design*, 147(6):061702 (2025). [Link](#)

- 2025 **Saeid Bayat** and James T. Allison. "A practical open-source approach to Model Predictive Control using the Legendre–Gauss–Radau pseudospectral method." *Software Impacts*, 100769 (2025). [Link](#)
- 2025 **Saeid Bayat**, Nastaran Shahmansouri, Satya R. T. Peddada, Alex Tessier, Adrian Butscher, and James T. Allison. "Can Graph Neural Networks Help Identify Promising Thermal Management System Architectures Among Vast Numbers of Possibilities?" *Journal of Mechanical Design*, (2025). [Link](#)
- 2025 **Saeid Bayat**, Nastaran Shahmansouri, Satya R. T. Peddada, Alex Tessier, Adrian Butscher, and James T. Allison. "Extracting Design Information From Optimized Designs of Power Flow Systems: Application to Multisplit Thermal Management System Configuration." *Journal of Mechanical Design*, 147(11):112001 (2025). [Link](#)
- 2025 **Saeid Bayat**, Nastaran Shahmansouri, Satya R. T. Peddada, Alexander Tessier, Adrian Butscher, and James T. Allison. "Multi-split configuration design for fluid-based thermal management systems." *Journal of Mechanical Design*, 147(2):021705 (2025). [Link](#)
- 2025 **Saeid Bayat** and James T. Allison. "Control Co-Design With Varying Available Information Applied to Vehicle Suspensions." *Journal of Dynamic Systems, Measurement, and Control*, 148(1):011013 (2026). [Link](#)
- 2024 Mohammad Sadman Sakib, D. Todd Griffith, Sanower Hossain, **Saeid Bayat**, and James T. Allison. "Intracycle RPM control for vertical axis wind turbines." *Wind Energy*, 27(3):202–224 (2024). [Link](#)
- 2023 **Saeid Bayat** and James T. Allison. "SS-MPC: A user-friendly software based on single shooting optimization to solve Model Predictive Control problems." *Software Impacts*, 17:100566 (2023). [Link](#)
- 2021 **Saeid Bayat**, H. Nejat Pishkenari, and H. Salarieh. "Observation of stage position in a 2-axis nano-positioner using hybrid Kalman filter." *Scientia Iranica*, 28(5):2628–2638 (2021). [Link](#)
- 2019 **Saeid Bayat**, Hossein Nejat Pishkenari, and Hassan Salarieh. "Observer design for a nano-positioning system using neural, fuzzy and ANFIS networks." *Mechatronics*, 59:10–24 (2019). [Link](#)

Manuscripts Under Review and In Preparation

- Under Review **Saeid Bayat** and Lei Zuo. "Control Co-Design and Performance Assessment of a Hybrid Semi-Submersible Wind Turbine Platform with Flap-Type Wave Energy Converters." Revised manuscript resubmitted to *Journal of Mechanical Design*; currently under review.
- Under Review **Saeid Bayat** and Lei Zuo. "Data-Driven Surrogate Modeling for Frequency-Domain Boundary-Element Analysis." Revised manuscript resubmitted to *Ocean Engineering*; currently under review.
- Under Review Vishnu Vijayasankar, **Saeid Bayat**, Nahian Masud, Lisheng Yang, Chris Qin, and Lei Zuo. "Dry-Lab Experimental Study of the Power Take-Off System for a Scaled Dual Hydrofoil Marine Energy Converter." Submitted to *IEEE/ASME Transactions on Mechatronics*; currently under review.

In **Saeid Bayat**, Lisheng Yang, and Lei Zuo. "Control Co-Design for Deployable Marine Preparation Energy Systems." Invited review article for *Nature Reviews Electrical Engineering*; manuscript in preparation.

In **Saeid Bayat** and Lei Zuo. "From Wave Energy to Hydrogen: Modeling, Optimization, and Power Management of Offshore Green Hydrogen Systems." Manuscript in preparation for submission to *International Journal of Hydrogen Energy*.

Peer-Reviewed Conference Publications

- 2026 **Saeid Bayat** and Lei Zuo. "Control Co-Design and Performance Assessment of a Hybrid Semi-Submersible Wind Turbine Platform with Flap-Type Wave Energy Converters." Accepted for presentation at the ASME IDETC/CIE 2026 (Paper No. IDETC/CIE2026-194389).
- 2025 **Saeid Bayat** and Lei Zuo. "Design of Experiments-Based Analysis of a Hybrid Wind-Wave Energy System With a Spar-Torus Combination." In *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)* (2025). [Link](#)
- 2025 Jerry Zuo, **Saeid Bayat**, and Jing Sun. "A Hybrid Wind-Wave Energy Converter with Structural and Functional Synergy: Concept Design and Simulation." In *Proceedings of the OCEANS 2025 Great Lakes Conference* (2025). [Link](#)
- 2023 Abbas Bataleblu, Vedant, **Saeid Bayat**, and James T. Allison. "Control Implementation Challenges of HI-MSAC: Hinge Integrated Multifunctional Structures for Attitude Control." In *Proceedings of the 46th Annual AAS Rocky Mountain Section Guidance and Control Conference* (2023).
- 2022 Yong Hoon Lee, **Saeid Bayat**, and James T. Allison. "Control co-design using a nonlinear wind turbine dynamic model based on OpenFAST linearization." In *Applied Energy Symposium: MIT A+B*, pp. 5–8 (2022).
- 2021 **Saeid Bayat**, Yong Hoon Lee, and James T. Allison. "Control Co-Design of Horizontal Floating Offshore Wind Turbines Using a Simplified Low-Order Model." In *Proceedings of the Wind Energy Science Conference* (2021).

Research Experiences

2024–Now **Postdoctoral Research Fellow, MaRIInE Lab, University of Michigan, Ann Arbor, USA**

As a Postdoctoral Research Fellow in the Marine Renewable Energy (MaRIInE) Laboratory, I conduct research on the design, optimization, and control of marine renewable energy systems, including wave energy converters, riverine energy converters, floating offshore wind turbines, and hybrid wind-wave platforms. My work integrates ocean engineering, control co-design, optimization, and data-driven modeling to improve energy production and system performance. I develop simulation and experimental frameworks, contribute to NSF, DOE, and ONR proposals, and mentor undergraduate and graduate students in marine energy research.

- 2019–2024 **Research Assistant, Engineering System Design Laboratory (ESDL), University of Illinois at Urbana–Champaign, Urbana–Champaign, USA**
 As a doctoral researcher at the Engineering System Design Laboratory, I worked on the control co-design and optimization of complex mechanical and energy systems, with a particular focus on floating offshore wind turbines. My research examined how system architecture and controller design interact under operational constraints. I developed model-based design frameworks that integrate optimization, control theory, and data-driven methods to enhance performance, reliability, and design efficiency. This work introduced surrogate and graph-based modeling approaches for design knowledge extraction and resulted in several peer-reviewed publications and collaborations across system dynamics, optimization, and intelligent design.
- 2015–2018 **Research Assistant, Nano Robotics Laboratory, Sharif University of Technology, Tehran, Iran**
 As a research assistant at the Nano Robotics Laboratory, I worked on the design and control of an XY nano-positioning robotic stage for precision mechatronic applications. I developed neural network, fuzzy logic, and hybrid observer-based controllers and implemented them on embedded hardware for real-time operation. I also designed graphical interfaces to support experimental testing and data acquisition. This work resulted in several peer-reviewed journal and conference publications and strengthened my expertise in robotics, mechatronic system design, hardware–software integration, and experimental validation.
- 2011–2015 **Research Assistant, Intelligent, Autonomous and Distributed Systems (IDAS) Laboratory, Iran University of Science and Technology, Tehran, Iran**
 As a research assistant at the IDAS Laboratory, I was part of a student team that designed and developed a bio-payload recovery CanSat, which won third place in a national competition. My responsibilities included mechanical design, sensor and actuator integration, and system-level testing under real operating conditions. This project provided early experience in robotics, autonomous systems, interdisciplinary teamwork, and practical design under strict space, weight, and environmental constraints.

Teaching Experiences

- 2025 **Guest Lecturer (1 lecture), NA 540: Marine Dynamics III, University of Michigan, Ann Arbor, USA**
- 2025 **Guest Lecturer (1 lecture), NA 499/599: Marine Energy and the Blue Economy, University of Michigan, Ann Arbor, USA**
- 2024 **Co-Instructor, Mechatronics, Industrial & Enterprise Systems Engineering, University of Illinois Urbana–Champaign, USA**
- 2023 **Guest Lecturer (4 lectures), SE 498: Control Co-Design, University of Illinois Urbana–Champaign, Urbana–Champaign, USA**
- 2023 **Teaching Assistant, Mechatronics, Industrial & Enterprise Systems Engineering, University of Illinois Urbana–Champaign, USA**
- 2023 **Teaching Assistant, Digital Control Systems, Industrial & Enterprise Systems Engineering, University of Illinois Urbana–Champaign, USA**
- 2022 **Guest Lecturer (4 lectures), SE 498: Control Co-Design, University of Illinois Urbana–Champaign, Urbana–Champaign, USA**
- 2017, 2018 **Teaching Assistant, Mechatronic Systems, Department of Mechanical Engineering, Sharif University of Technology, Tehran, Iran**

2014 **Instructor**, *MATLAB Software*, Department of Mechanical Engineering, Iran University of Science and Technology, Tehran, Iran

Student Mentorship

- University of Michigan Mentored **3 Ph.D. students** and **2 undergraduate researchers**; outcomes include 2 journal publications, 2 conference papers, and 1 journal manuscript currently under review.
- University of Illinois Mentored **2 Ph.D. students** and **1 undergraduate researcher**; outcomes include 1 journal publication and 1 conference paper.

Proposal Contributions

- 2026 **Lead Author & Senior Personnel**, *NSF EDSE Full Proposal: Exploring Aero-Hydro-Servo Coupling Dynamics of Offshore Energy Systems through Control Co-Design*, National Science Foundation (NSF EDSE)
Proposed budget: \$600,000. Submission Date: January 2026. Status: Under review. Main proposal-writing author; prepared and wrote most of the technical and proposal materials (~ 90% contribution).
- 2026 **Co-author**, *NSF TTP-T Full Proposal: Toward Autonomous Offshore Aquaculture Using Wave-Powered Energy Platforms*, National Science Foundation (NSF TTP-T)
Proposed budget: \$1.2 million. Submission Date: May 2026. Status: Under review. Contributed to proposal writing, technical development, and revisions (~ 45% contribution).
- 2026 **Co-author**, *NSF Full Proposal: Transforming Wave Energy Converter Array Design via Metastructure-Inspired Innovation*, National Science Foundation (NSF)
Proposed budget: \$600,000. Submission Date: March 2026. Status: Under review. Contributed to proposal writing, technical development, and revisions (~ 25% contribution).
- 2026 **Co-author**, *University of Michigan Strategic Initiative Fund: AI-Enhanced Experiential Training for Building U.S. Leadership in Ocean Energy and the Blue Economy*, University of Michigan Strategic Initiative Fund
Submission Date: February 2026. Status: Submitted. Contributed to proposal writing and development of technical and educational content (~ 45% contribution).
- 2025 **Lead Author**, *DOE EERE Full Proposal: WECSAM – Wave Energy Converter for Sustainable Aquaculture Monitoring*, U.S. Department of Energy (DOE EERE)
Proposed budget: \$2.5 million. Submission Date: February 2025. Status: Program terminated. Main proposal-writing author; prepared and wrote all technical and proposal materials (~ 90% contribution).
- 2025 **Co-author**, *NSF CPS-Medium Full Proposal: Collaborative Research: All-weather Coastal Environmental Monitoring and Communications through Hybrid Wind-Wave Energy Harvesting (Marinet)*, National Science Foundation (NSF)
Proposed budget: \$1.0 million. Submission Date: May 2025. Status: Under review. Contributed to technical writing, proposal development, and revisions (~ 35% contribution).
- 2025 **Co-author**, *ONR White Paper: Hybrid Floating Wave-Wind Platform for Resilient Power at Navy Facilities*, Office of Naval Research (ONR)
Proposed budget: \$19.15 million. Submission Date: August 2025. Status: Declined. Contributed to technical writing, proposal development, and revisions (~ 35% contribution).

Professional Service

Journal Reviewer *Applied Energy, Ocean Engineering, Journal of Mechanical Design, Structural and Multidisciplinary Optimization*

Leadership Hiring Manager, Marine Lab, University of Michigan

Professional Development

2025 Completed the Postdoctoral Short Course on College Teaching in STEM, Center for Research on Learning and Teaching (CRLT), University of Michigan, Ann Arbor

Computer Skills

Simulation Tools OpenFAST, WEC-Sim, Capytaine, WecOptTool, OpenMDAO, Dymos

Programming Languages C++/C, Python

Mathematical Analysis MATLAB (M-File, Simulink, GUI)

Machine Learning PyTorch

CAD Tools CATIA, SOLIDWORKS

Hardware STM Microcontroller, TI Microcontroller

Real-Time Control Speedgoat, TwinCAT, Beckhoff

Media Coverage

2026 **“Could Lake Michigan Waves Help Power Beaver Island?”**, *UpNorthLive*, June 24
Featured for University of Michigan wave energy research on Beaver Island. [Link](#)

2026 **“This Island in the Great Lakes Wants to Tap Waves for Energy”**, *Grist*, June 25
Featured in a news article on the University of Michigan's wave energy research. [Link](#)

2026 **“Michigan Island Needs Reliable Power. Can Great Lakes Waves Provide It?”**, *Bridge Michigan*, June 17
Featured and quoted for University of Michigan wave energy research. [Link](#)

2026 **“A Floating Platform that Captures both Wind and Wave Energy”**, *The Science Matters*, April 1
Featured for research on hybrid floating wind–wave energy systems. [Link](#)

2025 **“U-M Team Wins ASME Best Student Paper”**, *University of Michigan NA&ME*, September 19
Featured for receiving the ASME IDETC 2025 Best Student Paper Award. [Link](#)

References

- Dr. **Lei Zuo**
Herbert C. Sadler Collegiate Professor of Engineering
Professor, Department of Naval Architecture and Marine Engineering
Professor, Department of Mechanical Engineering
Director, NSF I/UCRC GO Blue Center
University of Michigan
2600 Draper Drive, Ann Arbor, MI 48109, USA
Phone: +1 (734) 660-9328
leizuo@umich.edu
<https://marine.engin.umich.edu/>
- Dr. **James T. Allison**
Jerry S. Dobrovolsky Faculty Scholar
Professor, Department of Industrial and Enterprise Systems Engineering
Affiliate Professor, Department of Aerospace Engineering
University of Illinois Urbana–Champaign
104 S. Wright Street, Urbana, IL 61801, USA
Phone: +1 (217) 244-4319
jtalliso@illinois.edu
<http://systemdesign.illinois.edu/home/>
- Dr. **Yong Hoon Lee**
Assistant Professor, Department of Mechanical Engineering
The University of Memphis
375 Dudley Building, 3710 Walker Ave, Memphis, TN 38152, USA
Phone: +1 (901) 6785004
yhlee@memphis.edu
<https://yonghoonlee.com/>